

Prediction Workflow

Geoeffective CME Prediction Using SOHO and SDO Observations

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1. Open the Package

Download `dnet_paper_examples.zip` from the accompanying Zenodo record. Extract the archive and open a terminal in the `dnet` directory.

File or directory	Purpose
<code>demo/dnet_best.pt</code>	Pretrained model
<code>demo/test.csv</code>	Event identifiers and labels
<code>demo/test_tensors/</code>	Prepared prediction inputs

2. Install Dependencies

Use Python 3.10 or later. Run:

```
python -m pip install -r requirements.txt
```

The program uses CUDA when available and otherwise runs on the CPU.

3. Run Prediction

```
python demo/test.py --num-workers 0
```

The program loads the pretrained model, reads the event list and prepared inputs, computes predictions, and evaluates the results.

4. Read the Results

The terminal displays each event's identifier, true label, predicted label, probability and classification outcome. Dates and paper figure references appear alongside the paper examples. Evaluation metrics follow the event results.

For training instructions and additional options, see `README.md`.